



# Intelligent product art design based on smart equipment and machine learning algorithm: practice effect and trend analysis

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## Abstract

With the development of the society, people's material needs are increasing rapidly. Based on this background, intelligent household furniture products with high-tech content are integrated into the field of interior design, and then gradually infiltrated into people's production and life. And from a smart technology perspective, we analyze the art design of smart home products based on machine learning algorithm. This algorithm can make the actual classification results of the test samples consistent with the network output values, and the error values can also meet the accuracy requirements, so as to effectively determine the defect types of the parameter samples. Among them, the art design of smart home products includes home equipment network group, embedded gateway implementation, cloud server construction and interaction design, independent product auxiliary control system and other parts. Finally, through the simulation test results of the intelligent positioning function of the product, we can know that the communication efficiency of the product basically meets the requirements. Interior design products are indispensable in people's home life, and play an important role in the whole indoor environment, such as art paintings, bonsai, woven art, etc. In addition, household appliances and lamps will also affect the aesthetic feeling of the whole interior design. This study tries to introduce machine learning technology in the field of interior art design to make it more intelligent, and a kind of effective algorithm design is completed.

**Keywords** Machine learning · Product art · Art design · Intellectualization

## 1 Introduction

Smart home products not only improve their technological level in technology, but also have greatly improved and improved the art and personalization (Kaveh and Vazirinia 2020). Nowadays, people's pursuit of convenient, efficient, energy -saving, environmentally friendly, and safe smart home products is becoming increasingly urgent. Smart home product art has become a product that confirms to the trend of the times (Bouchard et al. 2012). It not only has powerful functions, but also becomes more and more complete in terms of safety and automation performance. Thus greatly facilitating people's life (Nacer et al. 2017). Today, with the globalization of economic development, the indoor decoration industry, decoration industry and real estate developers

at home and abroad have seen the development prospects of smart home products, and incorporate smart home products into the home design and decoration industry, making the technology the current development of the furniture industry is the mainstream, and smart home technology has the characteristics of convenient, efficient, energy saving, environmental protection, security, etc. (Gentry 2009; Nef et al. 2012). Therefore, smart home products have become the goal of modern domestic home life. In the past, high-tech, intelligent, and information housing design will only be applied to large villas or mansion areas, but now the form of smart home design has begun to enter the lives of ordinary people (Kim et al. 2019).

## 2 Relevant work

In this paper, a method based on the combination of SVM and Markov model to predict the operating power of intelligent equipment is proposed. At the same time, the

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model is encapsulated as an API interface (Trappey et al. 2015; Xu et al. 2012). The IOS mobile phone client can directly call this interface to obtain the estimated operating power of the smart device, and can push the message of abnormal operating state of the device to let the user know the abnormal energy consumption of the smart device in real time (Seem 2007). The products designed in the literature realize a variety of network connection modes between the mobile client and the operating system and intelligent devices, including Bluetooth, hotspot, LAN and Internet connection. Through the design of distributed synchronization information storage, the information synchronization of IOS multiple clients is realized (Shiraz et al. 2012). At the same time, the client page is developed using Weex cross platform development technology. The code only needs to be written once and deployed on IOS and Android mobile platforms (Kumar et al. 2016). The solution is designed to configure the client cache mechanism, thus reducing the time required for the system to load pages. The literature designed a personalized smart home system, and introduced machine learning technology (Smirek et al. 2016). Through open HAB, the intelligent device management of the home gateway can be carried out, and the devices with different protocols can be compatible. In addition, it can also extract the system and analyze the family data, and make smart home devices to achieve personal comparison and experimental verification (Vijay and Banga 2017). The literature analyzes the characteristics of household data and draws the conclusion that the data have regularity, randomness and variance at the same time (Saunders et al. 2015). Therefore, the LSTM model of attention mechanism is designed to predict the regularity parameters of equipment status, and the Gru - InFCN model is designed for random classification and behavior recognition (Mekruksavanich and Jitpattanakul 2021). The literature has designed the model updating module, which can update the parameters of the model through different types and groups of data, and can combine the final results of the identification model with the updated data of the prediction model to achieve an intelligent and personalized control method (Barcelos Silva et al. 2020).

### 3 Common models and optimization algorithms of machine learning

#### 3.1 BP neural network model and improvement

The distribution of data not only affects the learning rate and the complexity of the network model structure, but also affects the accuracy of the training network. Therefore, in order to ensure the performance of neural networks, data

sampling must be carried out, usually using a simple linear transformation method. Because the activation function selected by BP neural network is SIGMOID function, whose value is [0, 1]. When the sample input is limited between [0.05, 0.95] or [0.1, 0.9], the change rate of SIGMOID function value is very high, which helps to shorten the network integration time. The data processing formula is as follows:

$$x_i = \alpha \frac{x_i - x_{j\min}}{x_{j\max} - x_{j\min}} + \beta \quad (1)$$

Among them, when  $\alpha$  takes 0.9, it can complete the conversion of sample data values in the range of [0.05, 0.95]. If  $\alpha$  takes 0.8, the sample data value is changed to the range of [0.1, 0.9].

This article analyzes and selects the wall thickness of the bumper plastic body.  $A$ , enhanced the height of the tendon  $H$ , mold temperature  $T$ , injection time  $K$ , pressure control  $P$ , and control time  $t$ , and  $I$  is the input feature of the network as the network.

$$x_i = \{A, H, T, K, P, t, I\}^T \quad (2)$$

The number of nodes of the output layer is determined according to actual problems, that is, the type of parameter design defect in the bumper. The modified PP bumper is mainly produced in injection molding process, so the surface of the finished product is prone to many defects such as welding seams, air points, pores, and deformation. At the same time, parameters such as structural design and processing technology of the bumper have a certain effect on its overall shape, hardness and strength. Such defects can be attributed to general defects of the bumper. Therefore, the number of network nodes of the network output layer is set to three. Among them, the type of defects is the surface defect F1 and bumper overall defect F2, and the design parameters of non-defective samples are defined as F0.

Generally, the value of  $\eta$  is within the range of [0, 0.8]. If the value of the  $\eta$  is large, the convergence speed of the network will be accelerated, but the corresponding pairs may cause the network to oscillate; if the value of the  $\eta$  is small, although the network will be stable, the convergence speed is slow. In addition, in the early days of network training, we hope that the adjustment of weights will be close to the convergence value as soon as possible, but in the later stage of training, we hope that network convergence is more stable. Therefore, the learning rate of the adaptive gradient decrease method is constantly changing during the formation of the network, which can improve the network performance:

$$\eta(k) = \begin{cases} 1.05\eta(k-1) & E(k) < E(k-1) \\ 0.7\eta(k-1) & E(k) > 1.04E(k-1) \\ \eta(k-1) & \text{other} \end{cases} \quad (3)$$

It can be seen from the formula that if the network connection weight and output layer standards are small, the positive correction and hidden layer standards will also be reduced, resulting in slower network training and defects. Therefore, the initial value of the weight and threshold of the output layer is generally randomly selected between  $-1$  and  $+1$ , and the initial value of the hidden layer is relatively small, such as:

$$w_{ij} = \gamma \frac{w_{ij}^*}{\sqrt{\sum_{i=1}^{n_1} w_{ij}^2}} \quad (4)$$

The most commonly used single-polar sigmoid function is used as the activation function:

$$f(x) = \frac{1}{1 + e^{-x}} \quad (5)$$

As shown in Table 1, three groups of three samples are randomly selected as test samples to enter the network model of test data and obtain:

It can be seen from Table 1 that the output value of the network is very consistent with the actual classification of the test sample, and the square error value basically meets the accuracy requirements of the sample parameters.

The optimal network model has  $w_j$ ,  $\beta_j$ ,  $\theta_j$  as:

$$\sum_j^m \beta_j f(w_j \cdot x_i + \theta_j) = t_i, i = 1, \dots, n \quad (6)$$

Then the above equation can be written as:

$$H\beta = T \quad (7)$$

At the beginning of learning, there is no need to adjust the hidden layer threshold  $\theta_j$ , then training the network is

equivalent to finding the solution of the linear system  $H\beta = T$  with least squares  $\beta$ :

$$\|H\beta' - T\| = \min_{\beta} \|H\beta - T\| \quad (8)$$

The above linear system has a minimum of minimum multiplied solutions:

$$\beta' = H^+ T \quad (9)$$

Among them,  $H^+$  is the broad sense of matrix  $h$ . Then this specific connection has the following characteristics:

Get an improved neural network model by training sample sets, and calculate the sensitivity of each random parameter based on the output of the network.  $W$  is the cause of product performance, where  $W = (\mu, \delta)T$  can be obtained from the network output. After the normalization, the output expression of the network is:

$$W^* = \sum_{j=0}^m \beta_j g(w_j \cdot x_i + b_j) \quad (10)$$

Through calculation, the design reliability indicator, design reliability, and the sensitivity of each random parameter are shown below:

$$\beta = (1.3504, 1.2361), R = (0.9115, 0.8907) \quad (11)$$

$$\frac{dR}{dX^T} = \begin{bmatrix} -5.417 \times 10^{-4} & -3.249 \times 10^{-4} \\ -0.746 \times 10^{-4} & 0.627 \times 10^{-4} \\ 1.614 \times 10^{-4} & 0.781 \times 10^{-4} \\ 0.489 \times 10^{-4} & -0.703 \times 10^{-4} \end{bmatrix} \quad (12)$$

### 3.2 Support vector machine model

If the sample input data are  $(x_i, y_i)$ ,  $i = 1, 2, \dots, N$ , where the input vector  $x \in \mathbb{R}^m$ , the corresponding output vector is  $y_i$ , and  $y = f(x)$  is the estimated output, then the linear approximation function of the equation can be expressed as Eq. 13:

$$y = f(x) = w^T \phi(x) + b, b \in \mathbb{R} \quad (13)$$

The maximum value of the function for  $a_i$  is:

$$Q(a) = \sum_{i=1}^n a_i - \frac{1}{2} \sum_{i,j=1}^n a_i a_j y_i y_j (x_i \cdot x_j) \quad (14)$$

Since the SVM algorithm of radial basis function (RBF) is superior to other kernel functions, its mathematical formula is:

$$K(x_i, x_j) = \exp \left[ -\frac{(x_i - x_j)^2}{\sigma^2} \right] \quad (15)$$

The formula of the selection variance of the training fitness function (MSE) is:

**Table 1** Part of neural network sample test results

| No | Actual output |    |    | Network output |        |        | Secondary error |
|----|---------------|----|----|----------------|--------|--------|-----------------|
|    | F0            | F1 | F2 | F0             | F1     | F2     |                 |
| 1  | 1             | 0  | 0  | 0.9811         | 0.0549 | 0.0647 | 0.0080          |
| 2  | 1             | 0  | 0  | 1.0106         | 0.1014 | 0.0250 | 0.0108          |
| 3  | 1             | 0  | 0  | 0.9939         | 0.0275 | 0.1118 | 0.0133          |
| 4  | 0             | 1  | 0  | 0.0004         | 0.9510 | 0.0134 | 0.0036          |
| 5  | 0             | 1  | 0  | 0.0027         | 0.9357 | 0.0141 | 0.0057          |
| 6  | 0             | 1  | 0  | 0.0189         | 0.9741 | 0.1078 | 0.0131          |
| 7  | 0             | 0  | 1  | 0.1156         | 0.0016 | 1.0347 | 0.0138          |
| 8  | 0             | 0  | 1  | 0.0320         | 0.1087 | 1.0046 | 0.0127          |
| 9  | 0             | 0  | 1  | 0.0099         | 0.0527 | 0.9879 | 0.0034          |

$$f = \frac{1}{K} \sum_{i=1}^K (y_i - \bar{y}_l)^2 \quad (16)$$

### 3.3 Construction of SVM and Markov combination model

The calculated predicted data and the actual data are divided into  $n$  intervals according to the current situation, which are represented by the symbol  $E_i$  ( $i = 1, 2, \dots, i$ ).

The probability that the error state  $E_i$  becomes the error state  $E_j$  after  $K$  steps is expressed by the following mathematical formula:

$$P_{ij}^{(k)} = \frac{s_{ij}^k}{s_i}, j = 1, 2, \dots, 3 \quad (17)$$

The matrix  $P(k)$  shows the transfer law between the system error states, and the Markov mathematical formula is:

$$\begin{aligned} M(1) &= M(0)P(1) \\ M(2) &= M(0)P(2) \\ &\dots\dots\dots \\ M(k) &= M(0)P(k) \end{aligned} \quad (18)$$

Assuming that the current data error prediction system has five error states, the prediction error at time 0 of the system is state  $E_2$ , then  $M_0$  is expressed as  $(0, 1, 0, 0, 0)$ , and the state transition probability matrix  $P(k)$  of  $M(k)$  confirms that if the fourth column of  $M(k)$  is the maximum value, then  $M(k) = (0, 0, 0, 1, 0)$ , it is considered that the next error prediction system is most likely to be in state  $E_4$ . According to the determined error state, take the average error value to correct the prediction result of SVM model:

$$Y = y \left( 1 - \frac{(A_i + B_i)}{2} \right) \quad (19)$$

The deployment process of the specific machine learning model is shown in Fig. 1.

In the offline part, the implementation of SVM algorithm adopts LIBSVM development package. First, format the sample data collected by the server into the data format required by LIBSVM. The data format specified by LIBSVM is:  $\langle \text{label} \rangle [\text{index1}]: [\text{value1}] [\text{index2}]: [\text{Value2}]$ .

In this paper, a week's data are selected as the sample data to train the SVM model. It can be seen from Fig. 2 that the overall prediction results are good, but there are errors in some time periods, which need to be corrected.

The learning rate of BP algorithm is 0.01, the support vector machine is SVR type,  $C$  is set to 80, and other parameters are taken as default values. Repeat the calculation for 50 times and take the average value. The

comparison results of three learning algorithms are shown in Table 2.

It can be seen from Table 2 that the improving neural network model estimates the constraint between the design parameters of the shaft cover and the constraint parameters. The equity error and deviation of the model result and the actual sample result are extremely small. The training results of these three methods are very close, but for the test set, the method of this article has significantly improved the BP neural network algorithm, and in terms of the promotion mode and the support vector the machine is equivalent, far exceeding the other two methods in learning.

## 4 Intelligent home product art design

### 4.1 Smart home products overall design

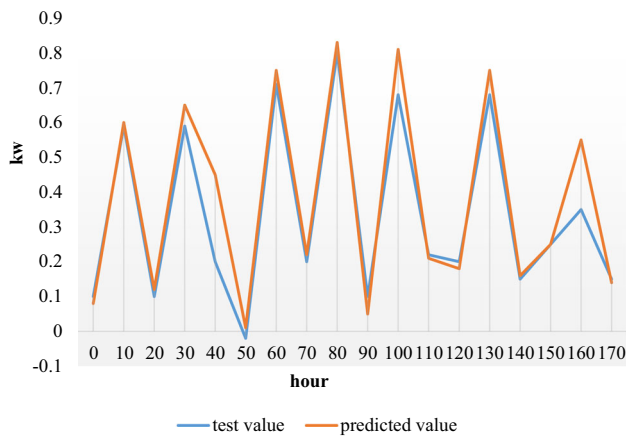
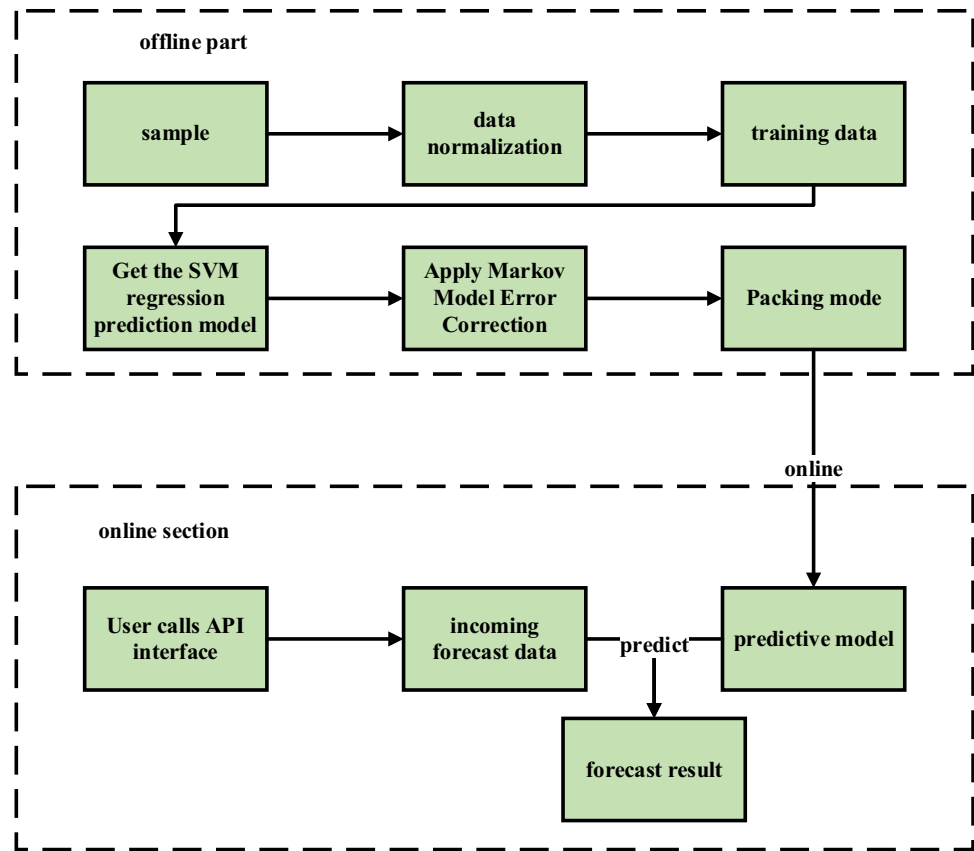
The artistic design of smart home products includes four parts: home equipment network, embedded gateway implementation, cloud server construction, and interactive design. Family equipment networking includes data such as collecting indoor and outdoor temperature and humidity, light intensity and other data, controlling indoor LED lights, and controlling air -conditioning temperature; embedded gateways include openHAB and binding OpenHAB devices; cloud server construction includes cloud server implementation, database interaction, machinery, machinery Learning model interface; the design of the interactive part includes the web part and the mobile application part. The Structure diagram of the overall system is shown in Fig. 3.

The system is divided into four parts: home equipment network, embedded gateway, cloud server and client.

The home appliance equipment network realizes communication between home appliances, sensor nodes and gateways. Because different protocol devices are encountered in network design, and the wireless network communication technologies of these devices are also different. At present, the most popular is WiFi, ZigBee and Bluetooth, but ZigBee, WiFi, and Bluetooth cannot be communicated.

Embedded gateway installation at home, complete the connection of terminal devices through the Internet network, send device status information, sensor data information, receive terminal sensor information and end user control commands; Control commands; connect family devices and family sensors through wireless networks, receive device status information and sensor data, and send device control commands.

The terminal server is implemented in the cloud. It is the core of the entire smart home control. The sensor data

**Fig. 1** Machine learning model deployment flow chart**Fig. 2** SVM fitting diagram

uploaded by the Internet network to receive the embedded gateway, the user's wearable smart sensor data and device

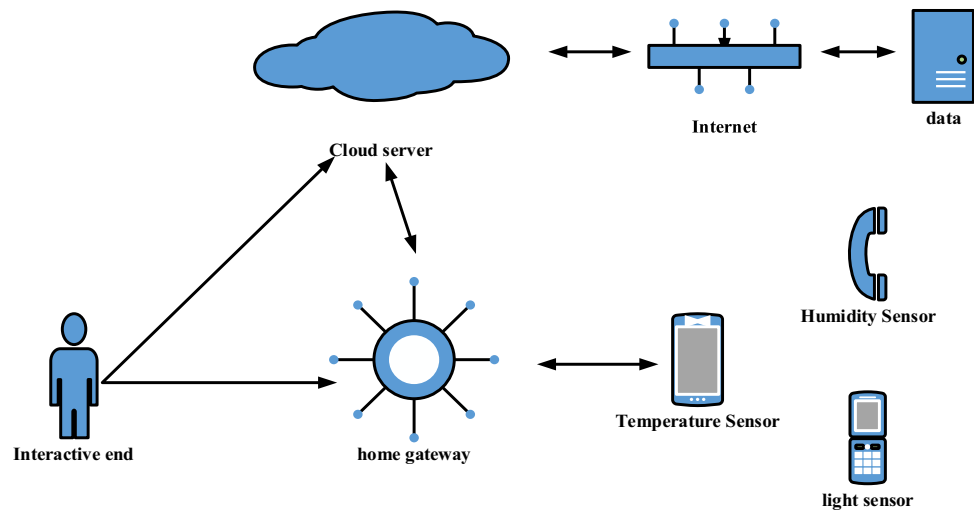
status information are used, and the received information is stored in the cloud database. Call the machine learning API to process, learn, predict, or judge the information, and send the device control command to the home gateway according to the prediction results or send an interactive information to the user. The machine learning API on the cloud server can be called by multiple parties.

System interaction, the system end of this design includes the web side and the mobile APP. The terminal completes the sending of the user control command and the receiving and display of the family environmental parameters. The sensor module associated with the mobile terminal can provide data to identify the user's behavior. By analyzing the behavior of data, the home equipment can be controlled according to the analysis results to achieve the system's "unspoken perception" interaction.

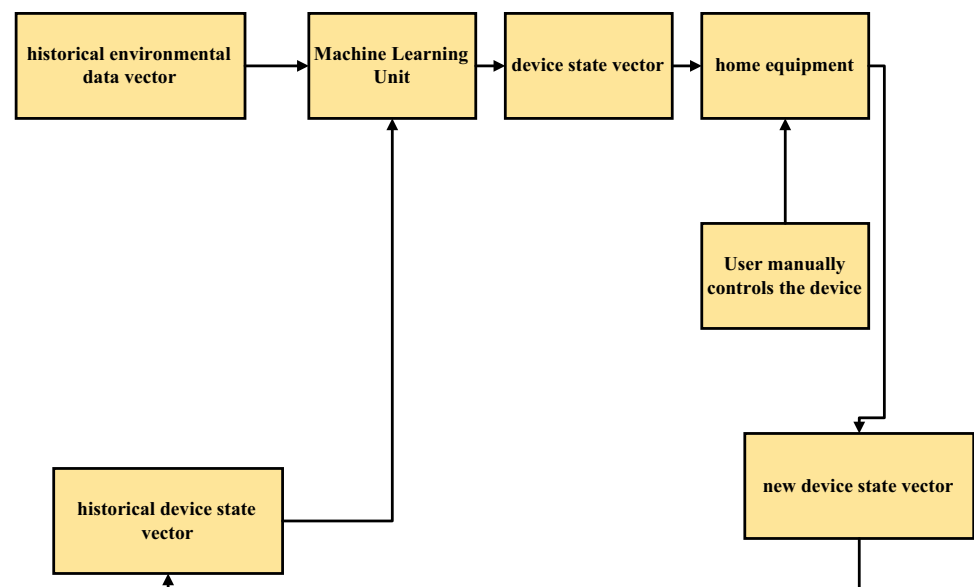
**Table 2** Comparison results of three learning algorithms

| Algorithm         | Total time consumption /s |        | Training set |        | Test set |        | Neuron number |
|-------------------|---------------------------|--------|--------------|--------|----------|--------|---------------|
|                   | Training                  | Test   | RMS          | Dev    | RMS      | Dev    |               |
| Improve algorithm | 0.5191                    | 0.1081 | 0.0207       | 0.1412 | 0.0153   | 0.1419 | 20            |
| SVM               | 28.649                    | 14.713 | 0.0155       | 0.1288 | 0.0137   | 0.1303 | 78            |
| BP neural network | 5.5338                    | 3.5219 | 0.0213       | 0.1401 | 0.0243   | 0.2512 | 20            |

**Fig. 3** Structure diagram of the overall system



**Fig. 4** Working process of auxiliary control subsystem



## 4.2 Product auxiliary control module design

Figure 4 shows the flow diagram of machine learning assistance system in smart home. As can be seen from Fig. 4, the system obtains the current working state, compares it with the data in the database, generates the device control instruction and transmits it, thus changing the working state of the device. This process is also constantly corrected during application. If there is a problem, the user can manually adjust the Settings, and this adjustment will be summarized into the system database, so that the system control can be corrected in the subsequent process.

## 4.3 Product intelligent regulation function simulation test

Some data of the family environment data set are shown in Table 3.

In Table 3, the visibility unit is km, the summary is the weather description, the apparent temperature is the temperature in the apartment (degrees Fahrenheit), the pressure is atmospheric pressure (Pascal), the wind speed unit is m / s, the time is the UNIX timestamp (in seconds), and the collection time of the environmental data set is 15 min.

The test time is November, and the time period is 8:00–20:00. In this time period, the environmental data are collected at a frequency of 300 s, and the equipment parameters are predicted and controlled using the



**Table 3** Partial data of environmental data set

| Visibility | Summai        | App are nt Temperature | Pressure | Wind Speed | Time          |
|------------|---------------|------------------------|----------|------------|---------------|
| 10         | Clear         | 28.23                  | 1026.97  | 9.04       | 1,466,140,644 |
| 9.88       | Mostly Cloudy | 27.63                  | 1026.39  | 9.81       | 1,466,144,280 |
| 10         | Clear         | 27.86                  | 1026.18  | 7.73       | 1,466,147,916 |
| 9.93       | Clear         | 28.84                  | 1026.15  | 6.18       | 1,466,151,552 |
| 10         | Mostly Cloudy | 27.69                  | 1025.74  | 7.86       | 1,466,155,188 |

equipment parameter prediction model. Record the following data every 300 s: outdoor temperature, air conditioning parameters and indoor temperature. After data processing, the experimental results shown in Fig. 5 are obtained.

#### 4.4 Product intelligent positioning function simulation test

In the internal positioning subsystem, the central controller first reads the positioning RSSI data packet sent by the network node of the positioning system, and judges the user's position in combination with the positioning engine. The working frequency of the radio frequency positioning module used in this system is 2.4 GHz, and the communication speed is set to 9600bps. The packet loss rates in the non-occluded environment and the occluded environment are tested, respectively. The results are shown in Tables 4 and 5.

The test results under barrier environment are shown in Table 5:

The barrier free environment test site is located on an open sports ground without partition in the middle. It can be seen from Table 4 that the RF system has higher communication efficiency and lower packet loss rate. The test site in the obstacle environment is located in the

experimental building with a wall in the middle. As shown in Table 5, when the transmission distance is 20 m, the packet loss rate is relatively low, with a maximum of only 1.2%. However, when the communication distance reaches 50 m, the packet loss rate increases significantly, with a maximum of 6.7%. Considering that the communication distance of about 20 m can meet the requirements in the ordinary family environment, the communication efficiency of the wireless radio frequency positioning system basically meets the standard.

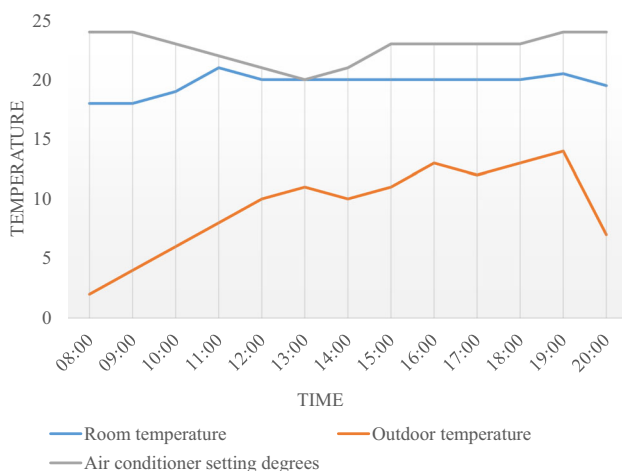
## 5 Development of interactive product art design under the background of artificial intelligence

### 5.1 Characteristic analysis of intelligent product art design

Practicality is one of the first aspects to be considered in intelligent design. Intelligent products without functional value are not good products. Not only keep the beauty and shape of the product, but also incorporate its practical value into the product design. When consumers choose smart products, they pay more attention to practicality, followed by the appearance and high-tech performance of the products. If the operation of an intelligent design product is complex, it will not be accepted by consumers no matter how beautiful it is. A product must be designed in combination with practicality and artistry. To judge whether a product is practical or not, the following reference features are available.

The birth of smart home technology is actually that people want to be more comfortable, convenient and safe at home. Therefore, the technical design is based on these needs and values. Without demand, there will be no new innovation. For example, consumers living alone tend to require products with safety systems, so they have higher demand for high-tech products that can provide them with safety protection than ordinary people.

When consumers and users use smart homes, their satisfaction is also affected by psychological factors, including “is it easy to use”, “is it used”, “is it appropriate”, “can it achieve the expected effect”, etc. These factors are



**Fig. 5** Indoor and outdoor temperature and regulation temperature change of air conditioner

**Table 4** Test results under barrier free environment

| Transmission distance /m | Packet length / byte | Sending cycle/ ms | Number of packets sent / piece | Number of accepted packets / piece | Packet loss rate / % |
|--------------------------|----------------------|-------------------|--------------------------------|------------------------------------|----------------------|
| 10                       | 64                   | 50                | 1000                           | 1000                               | 0                    |
| 10                       | 256                  | 50                | 1000                           | 1000                               | 0                    |
| 20                       | 64                   | 50                | 1000                           | 1000                               | 0                    |
| 20                       | 256                  | 50                | 1000                           | 998                                | 0.4                  |
| 50                       | 64                   | 50                | 1000                           | 995                                | 0.5                  |
| 50                       | 256                  | 50                | 1000                           | 991                                | 1                    |

**Table 5** Test results under barrier environment

| Transmission distance/m | Packet length/ byte | Sending cycle/ ms | Number of packets sent/piece | Number of accepted packets/piece | Packet loss rate/ % |
|-------------------------|---------------------|-------------------|------------------------------|----------------------------------|---------------------|
| 10                      | 64                  | 50                | 1000                         | 1000                             | 0                   |
| 10                      | 256                 | 50                | 1000                         | 1000                             | 0                   |
| 20                      | 64                  | 50                | 1000                         | 993                              | 0.9                 |
| 20                      | 256                 | 50                | 1000                         | 989                              | 1.2                 |
| 50                      | 64                  | 50                | 1000                         | 965                              | 4.2                 |
| 50                      | 256                 | 50                | 1000                         | 934                              | 6.7                 |

all from the psychological acceptance of consumers. A good smart home product can make users have a good psychological response, bring many benefits to consumers, and thus enhance the value of intelligent design products.

Intelligence can bring convenience to people, but too much intelligence cannot be imposed on product design to attract attention. Therefore, in the design of household products, the intellectualization should be appropriate, which should not only maintain its practicality, but also provide convenience for consumers. We need to understand the level of intelligence. It is time-consuming and laborious to exceed this level. We need to properly integrate intelligence into the design of household products. Any intelligent product must follow the principle of moderate intelligence. Only moderate products are good products. Intelligence mainly includes the following two aspects. The first aspect is to start with the practicality of the product. In order to improve the practical performance, intelligence is integrated into the design of intelligent products, rather than imposing some patterns on the products to improve profits. Science and technology are always making progress, but it is unrealistic to add intelligent elements to everything. Just as today's e-books cannot replace newspapers and paper books, intelligent design must be based on the real needs of users and the practicality of products, and excessive intelligence cannot affect the

user experience. The moderation principle in intelligent product design is very important. Only with moderate intelligence can we design more intelligent products.

## 5.2 Development path of intelligent product art design

Although technology is constantly developing, the ultimate subject of its service is people, so more humanized design can also become the mainstay of interactive art design. From the perspective of users, how to make a design or product more human must be analyzed from the perspective of users. By analyzing users' feelings when using products and interacting with artworks, the level of interaction itself can be improved.

If all things on the earth can interact intelligently, man-machine will gradually become the protagonist and carrier of future art creation, and a new art form will appear. This is man-machine art, which aims to express human needs. By using the nature of machines to plan people's social development and living conditions with automatic thinking, machines can also be used to determine the possibility of future imagination, thus breaking the pattern of artistic description and reality mapping. As a branch of new media art, interactive art design takes artificial intelligence technology as the media carrier, uses intelligent machine



language to convey artistic ideas and expressions, and then realizes the transformation of intelligent machine and human art.

Interactive art design spans many fields such as physics, biology and digital, and is a concentrated display form of art. At that time, art will no longer be a cold instrument. Small to nano robots and large to the earth can be set as interactive art works. The development of man-machine art not only needs to deal with the attribute of artificial intelligence as a practical machine, but also needs to conduct independent perspective and theoretical thinking through unlimited self-expression thinking. With the continuous development of artificial intelligence, the author believes that there are two major trends in the future: one is that artificial intelligence will be our friend in the future; Another trend is that artificial intelligence will become our enemy. What the author hopes to see is that in the future, we will gradually expand from the relationship of “unity of nature and man” between man and nature to the relationship of “unity of man and machine”, and realize the unity of self-subconscious.

## 6 Conclusion

The development of people's living standards will inevitably increase the material and spiritual needs, especially for the indoor residential environment, people will inevitably put forward new requirements. Based on this, more and more families are no longer interested in cheap products when choosing indoor furniture, but pay more attention to practicality and aesthetics. Therefore, the household products in the market are changing with each passing day. From the practical point of view, the popular products are often intelligent products, which are constantly developing towards a higher level and humanization in product style and art. Therefore, this paper carries out art design of smart home products based on machine learning algorithm. Through the research and analysis of relevant intelligent design, it opens up a new method in the field of art product design from the perspective of intelligent perception, improves the indoor environment and improves people's quality of life by combining intelligent design and product interior design.

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**Data availability** Data will be made available on request.

## Declarations

**Conflict of interest** The authors declare that they have no conflict of interests.

**Ethical approval** This article does not contain any studies with human participants performed by any of the authors.

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